Causal prediction under variable speed

Final validation study · 12-minute speaking target · FINAL_DATE_UNCONFIRMED

Can a compact causal model explain vibration efficiently—and leave interpretable residuals?

- Best measured validation configuration: lpv_L256_s17_a1
- Common-index MSE: 0.3612; independent test unavailable
- 200 separate fault-candidate records processed; physical fault accuracy unverified

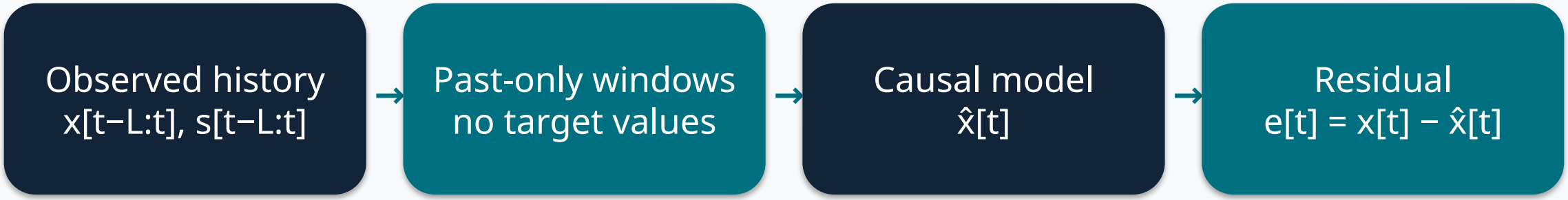
Data, provenance and limits

One reference trace and one validation trace; no independent test

- Train: 24,000 samples · Validation: 24,000 samples · fs = 3,200 Hz
- Provided x and spdnorm are used exactly; no record-wise rescaling
- Speed trajectory correlation: 0.999999571
- Speed units / reference health state unverified; normalized x is dimensionless
- Fault candidates: 200 separate 10-second records in 20 MAT parts
- sigtest.mat and sigfault2.mat not supplied

Causality before model complexity

Each target uses observed history only



Example: x[64] uses samples 0…63 only. One-step teacher forcing; no x[64] in model inputs.

The conventional comparison

AR/ridge and LPV-AR surrogate; no replicated Spa LPV-ARMA score

Ridge AR: minimize $\|y - Xb\|^2 + \alpha\|b\|^2$, $\alpha = 1$

- Zero, training-mean and persistence sanity checks
- OLS and ridge AR; ARX appends past speed and speed^2
- LPV-AR: lag coefficients vary polynomially with speed at t−1
- Measured ridge AR(256) validation MSE: 0.420058 (zero: 1.104077)
- LPV-AR surrogate has no ARMA innovation recursion; no exact paper reproduction

Three compact neural ideas

MLP lag mapping · GRU state · causal dilated TCN

MLP
lag vector → scalar

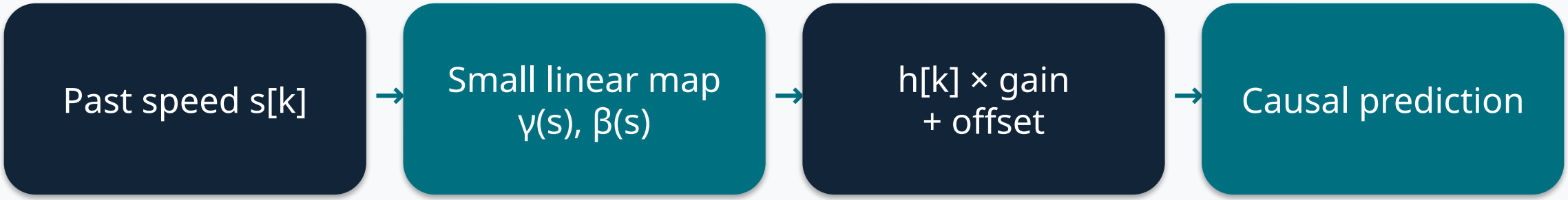
GRU
update / reset gates

TCN
causal dilations

Three separate model alternatives; compact networks, bounded epochs and recorded seeds. TCN: two kernel-3 convolutions per block; receptive field = 1 + 4 Σ dilation.

Speed-conditioning: contribution and controls

A project-level adaptation; not a world-first claim



Hypothesis: operating-speed information helps beyond extra capacity.
Controls: late concatenation, larger unconditioned TCN and off-line phase-shifted speed (negative control only).

Measured validation performance

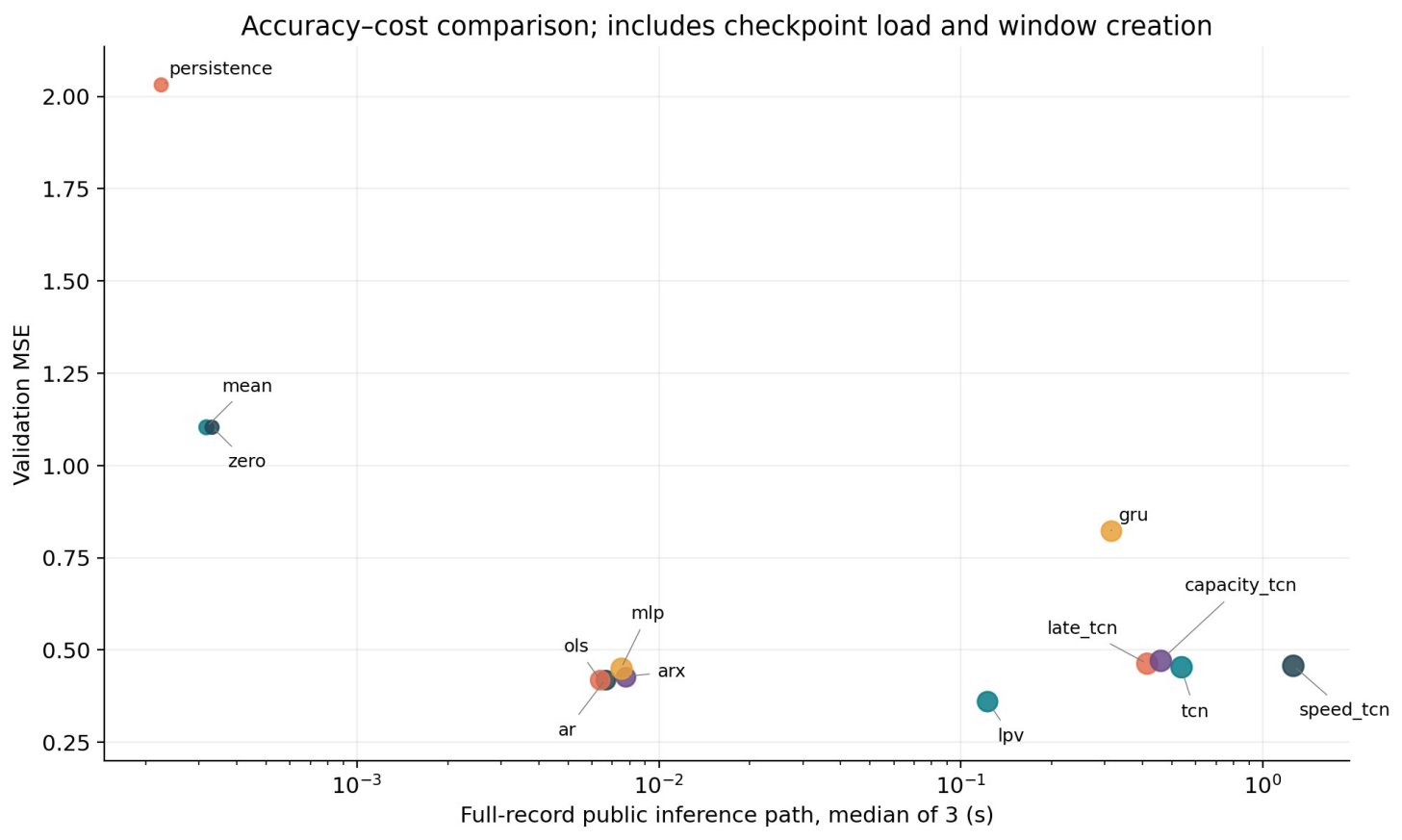
Every model evaluated at exactly t = 256…23999 (zero-based)

Model	Lag	MSE	Parameters	Train s	Record s
lpv	256	0.3612	769	0.766	0.123
ar	256	0.4201	257	0.069	0.007
ols	256	0.4201	257	0.264	0.006
arx	256	0.4278	259	0.064	0.008
mlp	64	0.4505	2113	1.490	0.007
tcn	128	0.4545	2425	26.812	0.536
speed_tcn	128	0.4582	2617	52.985	1.260
late_tcn	64	0.4633	2026	19.411	0.413

Best observed configuration per family; validation-selected. Timing includes checkpoint load, window construction and scoring. Full CSV retains every run.

Accuracy has a computational cost

CPU-only, measured public inference path; GPU benchmarks not run

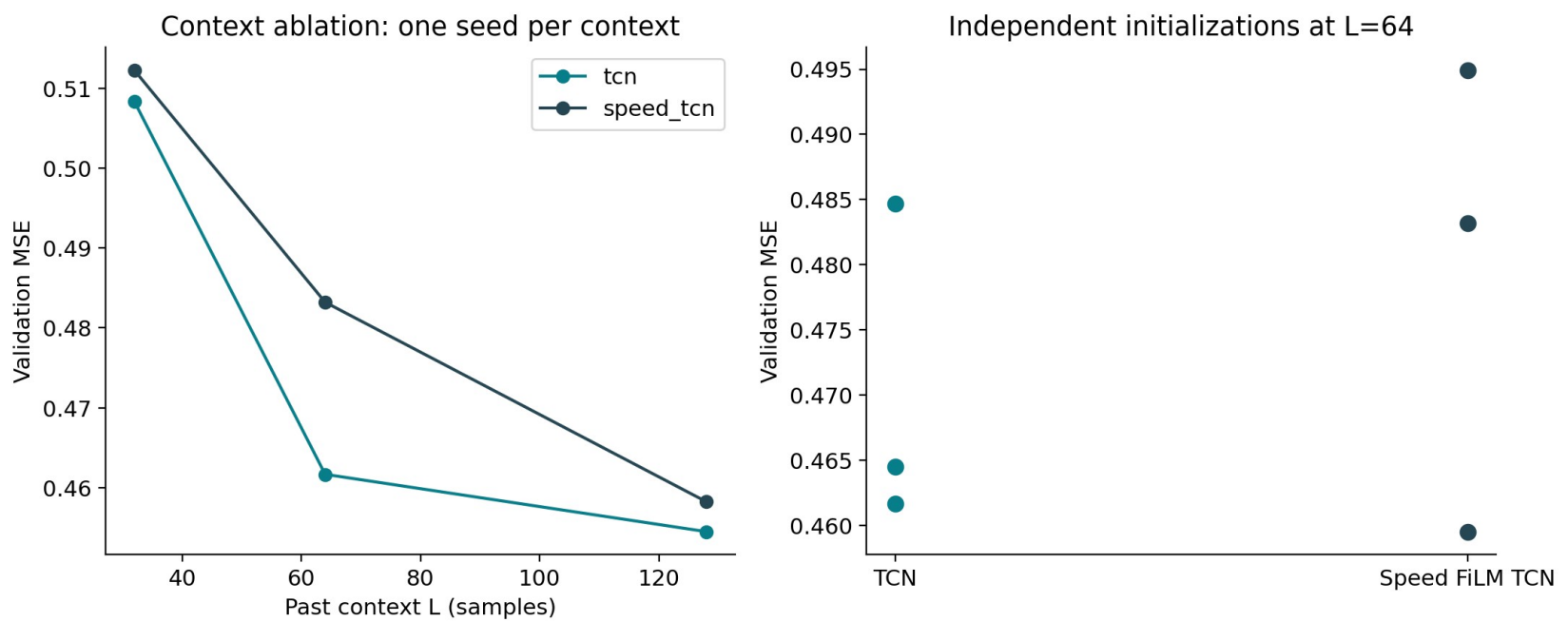


- Best model fit: 0.766 s
- Best model record path: 0.123 s
- Ridge AR(256): 0.0066 s / record
- Includes checkpoint load + windows; 3 repeats

Measured CPU wall time; details: reports/timing_and_hardware.md; no GPU measurements.

Does conditioning help consistently?

Context, seed and capacity controls; do not select only favorable seeds

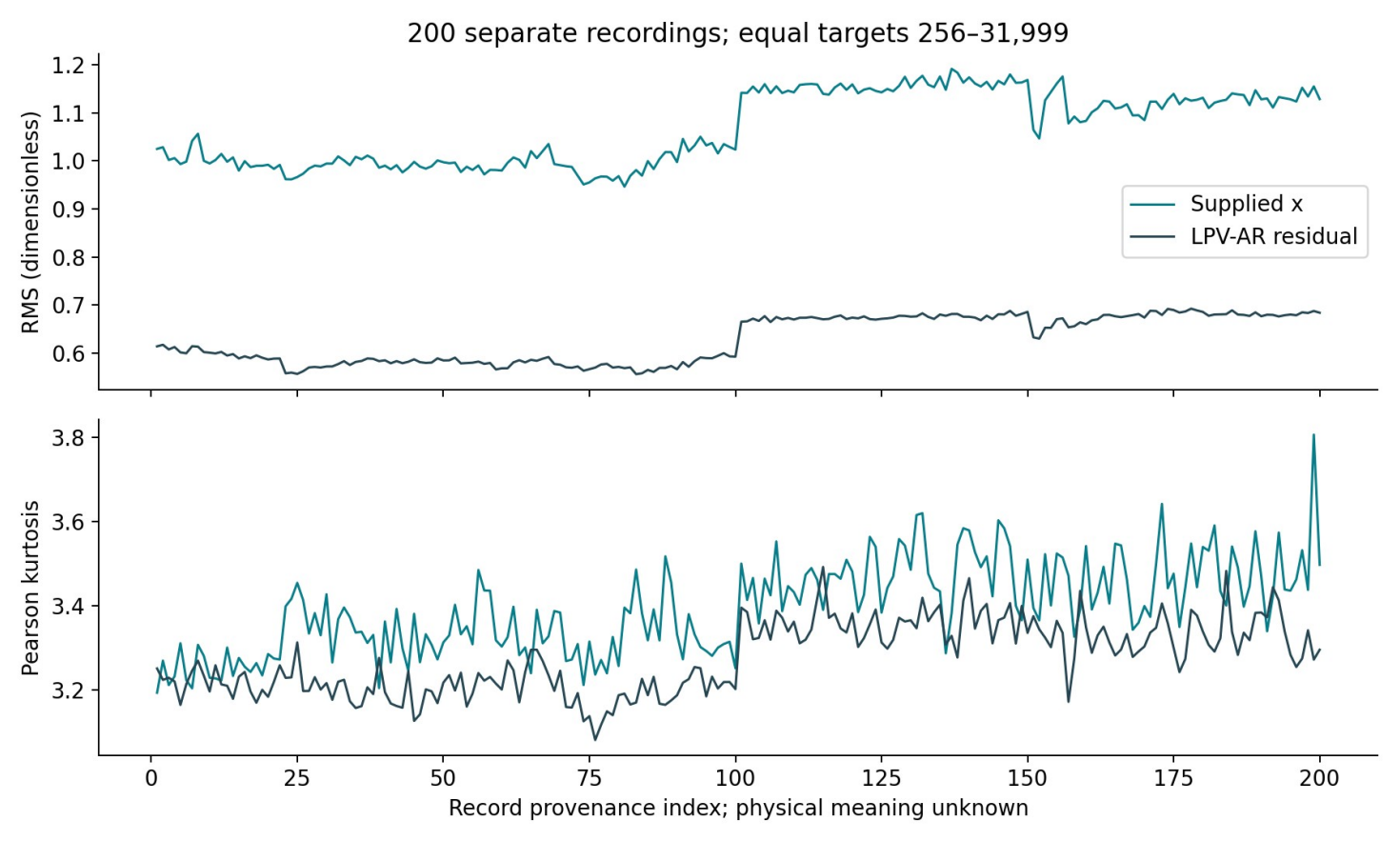


- speed_tcn: 0.4792 ± 0.0180 (3 seeds); tcn: 0.4703 ± 0.0125 (3 seeds)
- FiLM mean MSE: 1.90% higher
- Full controls: model_ablation.csv

Same validation targets for every context; points are initialization repeats, not independent recordings.

A residual does not prove a defect

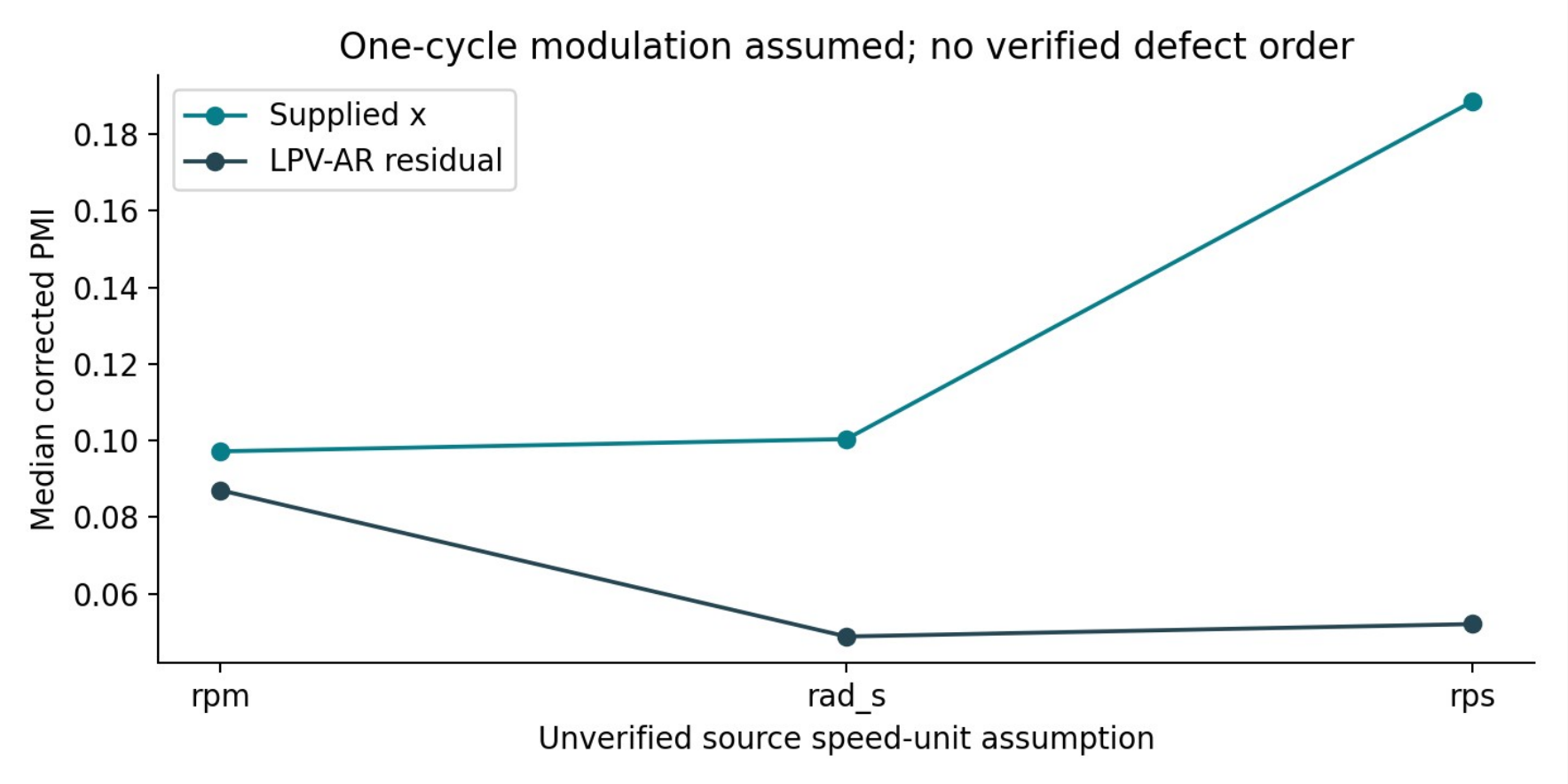
Prediction accuracy and diagnostic utility are distinct objectives



200 separate candidate records; equal 31,744-sample segments; dimensionless x/residual.

PMI and angular resampling need assumptions

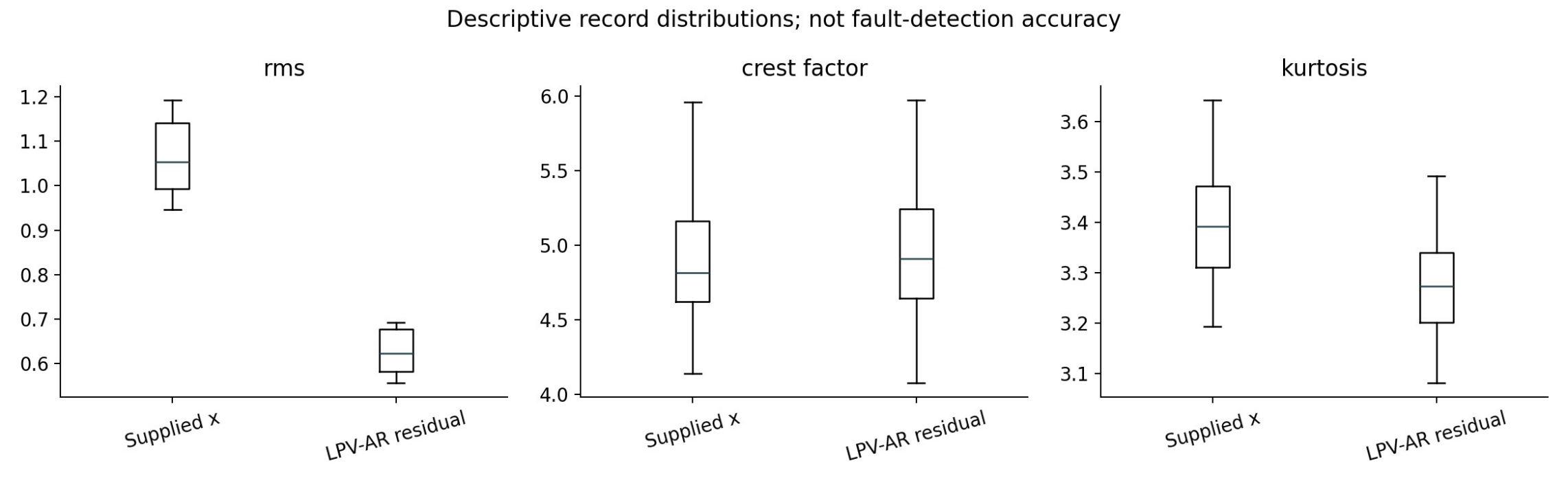
Explicit speed units; literal versus corrected MATLAB indexing



Actual fault-candidate descriptors; speed units / order are assumptions. Literal PMI lag T−1; corrected T.

Unlabeled records: measured descriptors

Record-wise raw/residual features; no invented health classes



Actual record-wise distributions; no verified classes. RMS / kurtosis / CK / PMI describe signals, not fault accuracy.

Limits of inference

Validation selection, one trace, repeated speed trajectory, unknown labels

- Only one short validation recording, reused for epoch/model selection
- Speed trajectories are almost identical: unseen-speed transfer untested
- Short neural training budget; rankings are budget-specific
- Physical speed unit / health labels unverified; patterns remain descriptive
- Spa LPV-ARMA exact comparison unavailable; final dates conflict in the PDF

What the evidence supports

Recommend using the measured table; validate on independent records next

- Measured best: lpv_L256_s17_a1, validation MSE 0.3612
- LPV-AR beats the tested neural models under this budget
- FiLM speed-TCN mean MSE is 1.90% higher than TCN (3 seeds)
- 200 separate fault-candidate records processed; physical fault accuracy unverified
- Next evidence: independent test and verified physical speed units / health labels

Sources and reproducibility

Editable decks generated from project outputs; exact CSVs and checkpoints retained

- Chen et al. (2025), MSSP 224, 112161. doi:10.1016/j.ymssp.2024.112161
- Bai, Kolter & Koltun (2018), TCN. Original author implementation accessed
- Perez et al. (2018), FiLM. Original author implementation accessed
- Full evidence status: research/literature_matrix.csv and source_access_log.md
- Rebuild: python scripts/build_presentations.py; inputs: reports/results_summary.csv